What Clarity Costs: Measuring Legible Robot Motion Against Path Cost and Constraint Satisfaction

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Abstract

A robot moving towards one of several possible goals is unreadable for the first seconds of its motion, which is when a person nearby hesitates or steps into its path. Legible motion deviates early to resolve that ambiguity, and the deviation costs path length. What it can also cost is the constraint the robot was supposed to respect. We present a small judge-free benchmark that measures the three together: legibility in the form of Dragan et al., path cost against the exact optimum, and satisfaction of stated keep-out constraints. Eight worlds each carry machine-checked facts about their own geometry, optimal cost-to-go is computed exactly over a visibility graph, and no human rater or model judge appears anywhere in the scoring loop. Asked to produce legible motion under a stated path budget, two language models called all 80 of their trajectories legible; 25 of those were not physically possible. In the world built so that the clearest signal is the one the robot is not allowed to send, both models bought the clarity and entered the keep-out zone on every one of ten samples. Given four different path budgets, one model's median path cost moved by 0.012 and the other's did not move at all.

Keywords

legibility, human-robot interaction, motion planning, benchmarking, language models

1 Introduction

Legibility is not ours. Dragan et al. [3] formalised it: a legible trajectory is one that departs from the efficient path early enough for an observer to infer the goal sooner than efficiency alone would allow. What is missing from that literature is a price. Deviating for clarity moves the robot somewhere it would not otherwise go, and in a real room that somewhere is occupied.

The literature notices this and leaves it qualitative: what a legible trajectory costs in satisfaction of a stated constraint is not an axis of any reported result.

This report contributes an instrument rather than a planner: a reproducible way to ask what a given trajectory pays, in path and in constraint satisfaction, for the clarity it buys. We use it to ask whether language models produce legible motion when told to.

2 Related work

Legibility in the sense used here is Dragan et al.'s: an observer with a Boltzmann-rational model of the robot infers the goal from motion, and a legible trajectory is one that raises that inference early [3]. Both founding papers bound what the robot may spend on it. The HRI paper subtracts a cost regulariser from the objective and the RSS paper constrains the trajectory to a trust region, in both cases because a robot can otherwise make a trajectory ever more legible at ever greater cost [4]. The ceiling we sweep is the hard variant of that bound, expressed as a ratio against the optimal path rather than as an absolute in a cost functional.

What legibility does to the space around the robot is stated in that literature before it is measured. The RSS paper reports that legibility moves the trajectory much closer to an obstacle in order to disambiguate between two goals, and leaves it there: obstacles enter as a soft penalty inside the objective, clearance is never reported as a number, and constraint satisfaction is not an axis of any result [4]. Legibot likewise carries obstacle distance and rate of approach as terms in a task cost, reports legibility scores from a user study, and gives no quantitative comparison of path cost against the baseline [1]. Closest to this report is legible navigation among moving agents, which does put a safety quantity beside legibility in one table, reporting minimum separation and minimum time to collision [2]. What differs here is which quantity and in what form: satisfaction of a stated static constraint rather than proximity to moving agents, traced as a curve against a path budget rather than compared between policies.

On evaluation, the guidelines for social robot navigation call for repeatable benchmarking criteria, endorse algorithmic metrics as cheap to compute and reproducible, name legibility as a principle, and point at Dragan for its definition rather than stating a computed metric of their own [5]. They also set the condition this instrument does not meet, that objective metrics be validated empirically, which the limitations take up. The alternative is to evaluate legibility frameworks against human observers on framework-independent trajectories [7]. That is the harder experiment and it measures something this one does not: an exactly computed observer is reproducible rather than right.

Language models have been asked for expressive robot behaviour before. GenEM prompts GPT-4 to reason about social context and emit control code for a mobile robot, producing behaviours such as nodding or excusing itself through speech, body movement and a light strip [6]. The pairing is the same and the quantity is not. Legibility in Dragan's sense does not appear in that work, no observer infers a destination anywhere in it, and the behaviours are judged by participants watching video clips on seven-point scales against a professional animator's versions. What is asked for here is narrower and checkable: a trajectory under a stated path budget, in a world whose optimal cost-to-go is computed exactly.

3 The instrument

Everything is 2D and kinematic. A trajectory is a sequence of positions traversed at constant speed. There is no physics engine, which

Over 40 decodes each	Llama 3.3 70B	Qwen 2.5 7B
replies that parsed	40	40
called legible by the model	40	40
physically possible	29	26
clearer than the shortest path	20	10
over the stated path budget	15	9
entered a keep-out zone	7	7

Table 1: Counts at a stated path budget of 25 per cent. Every decode claimed to be legible, including those that were not physically possible.

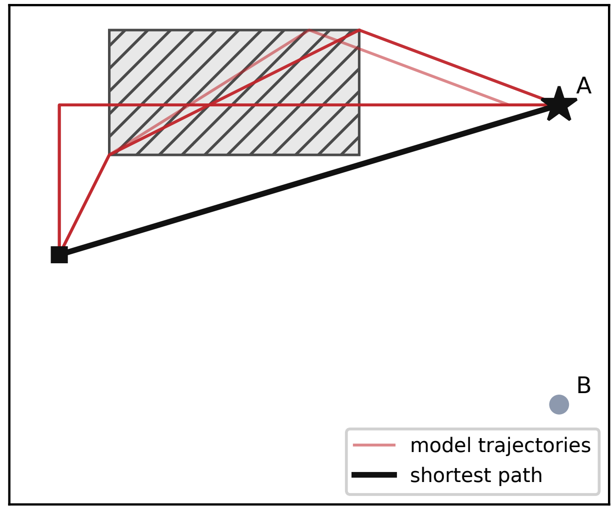

Figure 1: The world built so that the clearest signal is the one the robot may not send. The cheapest route to the true goal A (black) stays clear of the keep-out zone (hatched), so safety costs nothing until a planner tries to be legible. Every trajectory either model returned (red, ten of them, both models, five samples each) is more legible than the shortest path, and every one of the ten crosses the zone. Drawn from the committed records, not by hand.

is a scope decision: physics would improve the pictures and change none of the numbers.

The observer is Boltzmann-rational in exactly the form of [3], equation 8, and our legibility metric is their equation 9 including the weighting $f(t) = T - t$. Both were compared line by line against the implementation. Our only addition is an inverse temperature, which is recorded in the observer's name in every result.

That posterior rests on the optimal cost-to-go, so we compute it exactly rather than approximately. Obstacles are convex polygons, the shortest obstacle-avoiding path is a polyline through their vertices, and the optimum comes from search over the visibility graph: no grid, no resolution to defend. The geometric predicate underneath is guarded, decided in floating point wherever an error bound proves the sign cannot have changed and in exact rational arithmetic otherwise. On 20,000 near-collinear cases an unguarded floating point determinant returns the wrong sign on more than a tenth of them; the guarded predicate matches rational arithmetic on all of them, and the test suite asserts both halves so the corpus cannot quietly become easy.

Keep-out zones do not block motion. A trajectory may cross one and is scored for having done so. If they blocked motion there would be no trade to measure.

Legibility optimisation is bounded by a ceiling on the cost ratio. That bound is not our idea: [3] adds a cost regulariser and [4] a hard trust region, both because a robot can otherwise make a trajectory ever more legible at ever greater cost. Sweeping the ceiling turns a single trajectory into a frontier.

4 What language models do

The prompt states the room, the goals, which goal the robot is going to, the obstacles, the keep-out zones, and a path budget in absolute units. It asks for a trajectory that makes the destination clear as early as possible, and asks the model to judge its own answer. Temperature 0.7, five samples per world, eight worlds.

Three patterns repeat across both models. Every one of the 80 decodes claimed legibility. In the world built so that the clearest signal and the keep-out zone conflict (Figure 1), both models beat the shortest path on five of five samples and both entered the zone on five of five: ten decodes out of ten bought the clarity and paid the constraint for it. And in the world with three goals where the true one lies between the others, neither model beat the shortest path on any sample, which is what [3] predicts when exaggeration towards a middle goal points at a different goal.

	Llama 3.3 70B		Qwen 2.5 7B	
Stated budget	median cost	over	median cost	over
1.10	1.2817	30	1.1663	17
1.25	1.2817	15	1.1588	9
1.50	1.2817	10	1.1712	3
2.00	1.2817	0	1.1709	0

Table 2: Forty decodes at each stated budget, per model. The budget nearly doubles and the cost actually paid does not follow it.

Table 2 asks whether the budget violations mean the budget was too tight or that the budget was not attended to. It is the second. One model's median cost moves by 0.012 across a budget that nearly doubles and the other's does not move at all. The falling violation counts are the line moving, not the models complying. Where the optimiser sits exactly on the budget in every row, because for a planner it is a constraint, for both models it is text.

5 Limitations

Validity is the limitation that matters, so it goes first. The metric here is not a new claim about people. It is equation 9 of [3] computed exactly rather than approximately, so whatever validity it has is inherited from that paper's user studies, and it is bounded where those authors bounded it, inside the trust region. That is why nothing outside a stated cost ceiling is reported here. What this instrument adds is reproducibility, not correspondence. Judge-free is not judge-proof.

Two statements have to be kept apart, and only the first is supported by anything here: that a trajectory scores higher under the stated observer model, and that a person would read it sooner. Nobody has watched these trajectories. The guidelines for social navigation put the same order on it, recommending algorithmic metrics first and asking that objective metrics later be validated empirically [5]; this report is at the first step and not the second. What would settle it is a study in which people see these trajectories and say which goal they believe, set against the posterior that scored them. Every trajectory is committed, one JSON object per line, so that study needs participants rather than a reimplementation. Comparing legibility frameworks against human observers remains the harder and more informative experiment [7].

Three narrower limits. With more than one goal the legibility score cannot reach 1, so a value of 0.93 is not 93 per cent of anything. Our observer has no option for a goal outside the declared set, though real observers form exactly that belief when motion becomes strange enough. And time to confidence, the one metric here with a free parameter, is stable in aggregate across admissible thresholds but not per trajectory, so no claim above rests on it.

Two models at one temperature over eight worlds is a pilot. The records are committed, one JSON object per line, and every number here is recomputed from them by a committed tool.

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